


**UAECO5004CR1**

CIA 2 Assignment

# Fiscal and Monetary Policy Analysis of Germany (2015–2022):

## *Expansion, Contraction, Investment Trends, and the IS-LM Perspective*

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# 1.Introduction

Germany serves as an ideal case study for analyzing fiscal and monetary policy interactions due to its economic significance, historical commitment to fiscal discipline, and unique monetary policy framework under the European Central Bank (ECB). This paper aims to analyze Germany's economic fluctuations influenced by both internal and external factors by focusing on the fiscal and monetary policies from 2015 to 2022, focusing on periods of expansion and contraction. Specifically, we examine whether the government's fiscal decisions and the European Central Bank's monetary policies were expansionary or contractionary during different phases of the period.

The selected period of 2015 to 2022 is significant due to major economic shifts. Germany has historically adhered to a balanced budget policy, which aimed to prevent excessive government debt. From 2015 to 2019, Germany consistently ran budget surpluses, reducing its debt while maintaining strong economic growth. However, the COVID-19 pandemic in 2020 led to a sharp policy reversal, with the government adopting expansionary fiscal measures to support businesses and households. This shift from austerity to large-scale stimulus spending provides a unique opportunity to study the effectiveness of expansionary policies in a country known for its fiscal conservatism. With analysis of Germany's policy responses to economic shocks—such as COVID-19 stimulus measures and post-pandemic adjustments—the IS-LM model helps assess whether these interventions achieved their intended objectives or resulted in crowding out.

By integrating fiscal and monetary policy analysis with IS-LM modeling, this research provides a comprehensive evaluation of Germany's economic strategy during a transformative period, offering insights into the effectiveness of its macroeconomic policies.

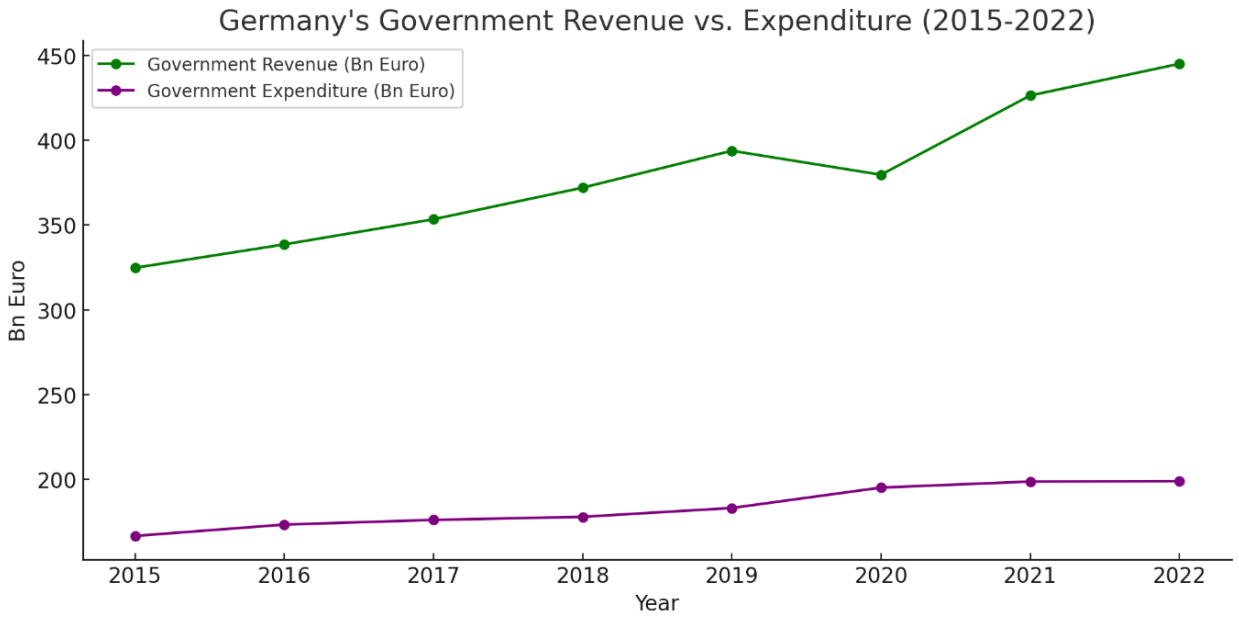
# 2. Budget Analysis

## 2.1 Overview of Budget Trends

| YEAR | GOVT. REVENUE (BN EURO) | GOVT. EXPENDITURE (BN EURO) | BUDGET SURPLUS/ DEFICIT(%) | PUBLIC DEBT TO GDP% |
|---|---|---|---|---|
| 2015 | 324.796 | 166.7626 | 1 | 71.2 |
| 2016 | 338.56875 | 173.448475 | 1.2 | 68.3 |
| 2017 | 353.39525 | 176.202625 | 1.3 | 64 |
| 2018 | 372.01775 | 178.024325 | 1.9 | 60.8 |
| 2019 | 393.77675 | 183.229775 | 1.5 | 58.7 |
| 2020 | 379.65275 | 195.261225 | -4.3 | 68 |
| 2021 | 426.33325 | 198.81735 | -3.6 | 68.1 |
| 2022 | 445.025 | 199.076925 | -2.5 | 65 |

As observed in the above table, Germany's government revenue increased steadily from €324.8 billion in 2015 to €445.0 billion in 2022. This growth can be attributed to tax collection mechanisms, economic expansion and rising income levels in the country.

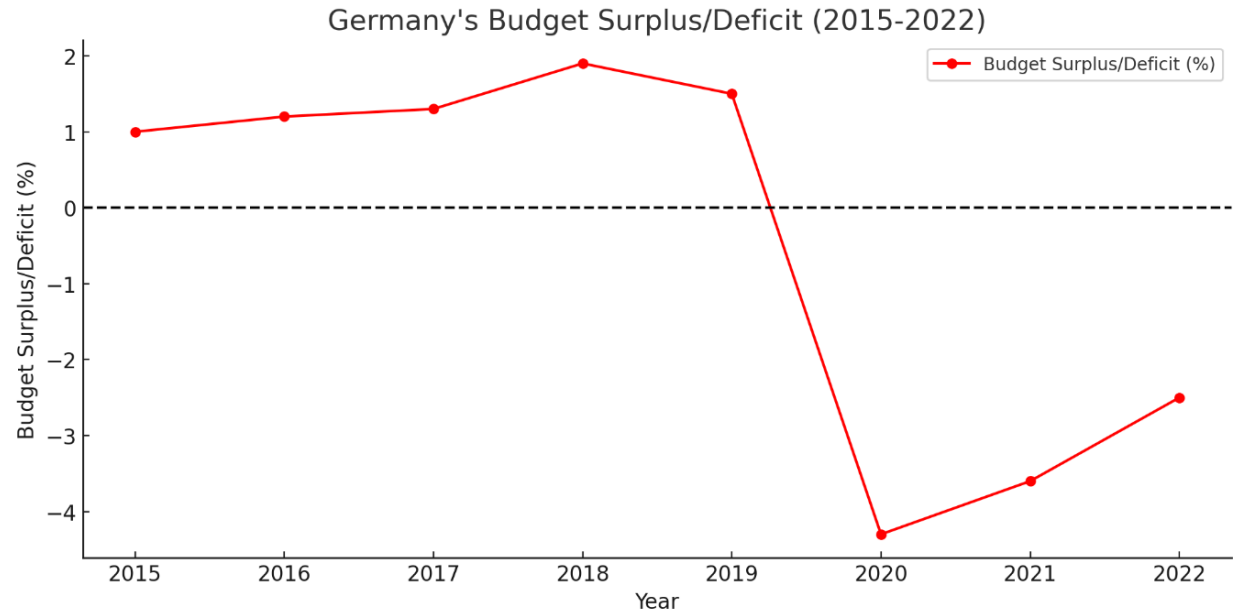
In a similar way, government expenditure also grew from €166.8 billion in 2015 to €199.1 billion in 2022. However it must be noted that the rate of expenditure growth remained moderate until 2019 after which it grew steadily. This was due to timely interventions and policies by the government during the Covid-19 crisis.




## 2.3 Budget Surplus and Deficit Analysis

It is evident that Germany maintained a budget surplus between the years of 2015-2019, with the surplus-to-GDP ratio peaking at 1.9% in 2019. This fiscal discipline was beneficial as it contributed to debt reduction and economic resilience. On the other hand, the onset of the pandemic in 2020 prompted a shift to deficit spending, as it fell to -4.3% of GDP in 2020, followed by -3.6% in 2021 and -2.5% in 2022. The deficits are reflective of substantial government support measures, including healthcare spending, social programs and stimulus packages.

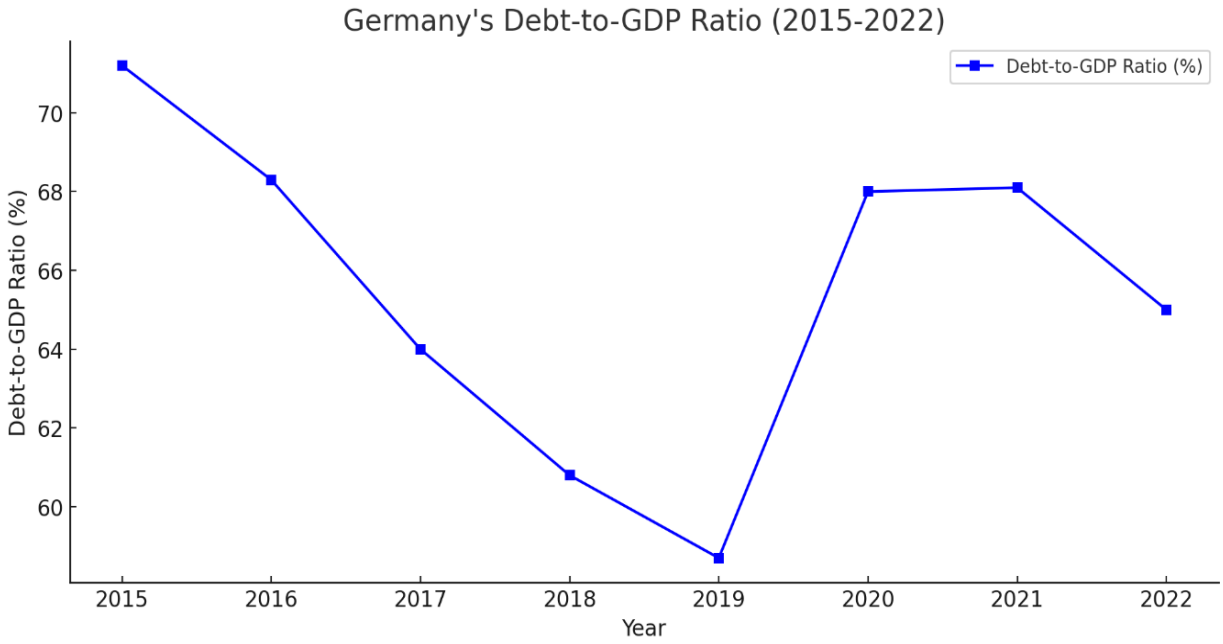
The shift from surplus to deficit also highlights the counter cyclical nature that fiscal policy seems to exhibit. Even though surpluses during the periods of economic stability contributed to debt reduction, economic downturns influenced deficit spending which was required to provide necessary fiscal support to sustain demand and prevent economic contraction.



### 2.4 Public Debt-to-GDP Ratio Trends

Germany's public debt-to-GDP ratio declined steadily from 71.2% in 2015 to 58.7% in 2019. This decline was due to budget surpluses and strong GDP growth. However, the fiscal response to the pandemic reversed this trend, with the ratio increasing to 68% in 2020 and stabilizing at 65% in 2022.

The increase in debt during the pandemic reflects increased government borrowing to finance economic stimulus programs. Even though Germany maintained a relatively low debt-to-GDP ratio compared to other advanced economies, the post-pandemic period required careful fiscal management to balance recovery with debt.




**3.**

# Methodology

### 3.1 Data Sources

This paper draws on multiple data sources to analyze Germany's fiscal and monetary policies and their effects on private investment from 2015 to 2022. The key sources include:

- Germany's Budget Reports and Revenue Data:
  Published by the Bundes finanz ministerium (Federal Ministry of Finance) and Deutsche Bundesbank (Germany's Monetary Authority), these reports provide data on government expenditures, tax revenue, budget deficits/surpluses, and fiscal measures.

- European Central Bank (ECB) Monetary Policy Data:
  Since Germany is part of the EU, the European Central Bank's(ECB) monetary policies play a crucial role. Key indicators include:
    - ECB policy interest rates (main refinancing rate, deposit facility rate).
    - Macroeconomic Indicators:
      GDP growth, inflation, private sector investment, and interest rate trends are sourced from IMF, World Bank, and Deutsche Bundesbank to analyze the broader macroeconomic environment.

## 3.2 Analytical Approach

To evaluate Germany's fiscal and monetary policies, policy changes between the years of 2015-2022 are to be studied before analyzing their impact on private investment. Additionally, this paper integrates economic theory and empirical analysis, using the IS-LM framework, correlation analysis, and crowding-out assessment.

### 3.2.1 IS-LM Framework

The Investment-Savings (IS) and Liquidity Preference-Money Supply (LM) model provides a theoretical basis for analyzing the interaction between fiscal and monetary policy. The study examines:

- Shifts in the IS curve: Expansionary fiscal policies (such as increased government spending during COVID-19) are expected to shift the IS curve rightward, increasing output and income. Contractionary fiscal policies (post-COVID deficit reduction) should shift it leftward.
- Shifts in the LM curve: Expansionary monetary policies (low ECB interest rates) should shift the LM curve rightward, reducing borrowing costs and stimulating investment.

### 3.2.2 Crowding-Out Effect Assessment

This paper evaluates whether Germany's fiscal policies crowded out private investment by increasing government borrowing and raising interest rates. Key indicators include:

- Debt-to-GDP ratio trends: Analyzing whether rising government debt led to higher borrowing costs for private firms.

- Bond yield movements: Examining German bond yields to assess whether higher deficits caused a rise in borrowing costs.
- Private sector investment response: Comparing trends in gross fixed capital formation with government borrowing patterns.

### 3.3 Justification of Approach

By integrating theoretical assumptions(IS-LM) and policy evaluation (crowding-out effects), this methodology ensures a comprehensive assessment of Germany's fiscal and monetary trends. The combination of quantitative data and theoretical models allows for a robust evaluation of policy impacts on private investment.

# 4. Germany's Fiscal and Monetary Policies

### 4.1 Fiscal Policy

| Period | Policy type | Key Indicators |
|---|---|---|
| 2015-2019 | Mild Contractionary (Fiscal consolidation) | - Budget Surplus & Schwarz-Null Policy: No net borrowing, keeping budgets balanced.<br>- Public Investment (€44 billion): Infrastructure (€17 billion), social services, and education.<br>- Debt-to-GDP Reduction: Dropped from 71.5% (2015) to 58.7% (2019).<br>- Minimal Crowding Out of Private Investment. |
| 2020-2022 | Expansionary ( Covid-19 response) | - €130 Billion Stimulus Package (2020): Direct financial aid to individuals & businesses.<br>- Kurzarbeit Program Expansion: Wage compensation for reduced work hours. |

| | | |
|---|---|---|
| | | - VAT Reduction (2020): From 19% to 16% to stimulate consumption.<br>- Sharp Increase in Government Spending: From €183.2B (2019) to €199.1B (2022).<br>Debt-to-GDP Surge: Rose from 58.7% (2019) to 72.25% (2021), then 65% (2022).<br>Budget Deficits: -4.3% (2020), -3.6% (2021), -2.5% (2022). |

From 2015 to 2022, Germany's fiscal policy was split into two phases: budgetary surpluses and fiscal consolidation (mild contractionary fiscal policy) from 2015 to 2019, and government spending to combat COVID-19 from 2020 to 2022. From 2015-2019, unemployment fell to 4.6%, the debt-to-GDP ratio dropped to 71.2% from 74.5%, and €44 billion was allocated to boost economic capacity, including €17 billion (0.6% of GDP) for infrastructure development (EuroStat, 2024). This approach balanced short-term growth with long-term fiscal stability while maintaining European fiscal targets (IMF, 2015). From 2016 to 2019, Germany stuck to its Schwarz-Null (Black Zero) policy, avoiding net borrowing. This caused the debt-to-GDP ratio to drop from 71.2% in 2015 to 58.7% in 2019 while ensuring stable economic growth and minimizing crowding out of private investment (Destatis, 2020). Public investment in social services, education, and infrastructure complemented fiscal prudence, contributing to steady economic sustainability.

To combat the COVID-19 epidemic, Germany's fiscal policy was drastically changed between 2020 and 2022. Instead of adopting the country's typical Schwarze Null balanced budget strategy, the country implemented expansive fiscal policy initiatives meant to stimulate the economy. Government expenditure saw a sharp increase, rising from €183.2 billion in 2019 to €199.1 billion in 2022. Direct financial assistance for people and enterprises that lost their incomes as a result of the quarantine was part of the €130 billion stimulus plan that the German government unveiled in 2020. Additionally, Germany declared that the Kurzarbeit program would be expanded. Under Kurzarbeit, workers receive full compensation for the hours they work and 60% of their wage for the hours they do not work. Furthermore, the expansionary fiscal policy implemented a temporary reduction in the value-added tax (VAT) from 19% to 16% to stimulate consumer spending.




*Source: EUROSTAT (2024)*

Nevertheless, these steps to combat the COVID-19-induced economic slump resulted in a significant rise in public spending, and in 2022, the total public budget deficit reached €127.3 billion. This resulted in budget deficits of -4.3% in 2020, -3.6% in 2021, and -2.5% in 2022 (Statistisches Bundesamt, 2024). Also, Germany's debt-to-GDP ratio increased as a result, from 58.7% in 2019 to 68% in 2020, reaching a peak of over 72.25% in 2021 before dipping little to roughly 65% in 2022 as seen above. The German government maintained a strong debt-to-GDP ratio sitting at 62.8% as of 2024. Germany's expansionary fiscal policy played a key role in bolstering employment, reducing the economic impact of the epidemic, and paving the road to economic recovery.

## 4.2 Monetary Policy

### 4.2.1 Monetary Policy Framework Overview

Germany's monetary framework consists of the Eurosystem which includes the ECB together with the national central banks of the Eurozone member states such as the Deutsche Bundesbank. The primary goal of the Eurosystem's monetary policy focuses on maintaining price stability through a medium-term annual inflation rate of 2% according to the Harmonised Index of Consumer Prices (HICP). The Bundesbank applies statistical projections, economic research and financial market knowledge to support the development and execution of monetary policy decisions within the ECB's Governing Council.

| Period | Policy Type | Key indicators |
|---|---|---|
| 2015-2019 | Expansionary | - Low Interest Rates: Deposit facility rate at -0.5% to encourage bank lending.<br>- Asset Purchases (PSPP): Injected liquidity into financial markets.<br>- TLTROs: Boosted bank lending to businesses and households. |
| 2020-2022 | Expansionary | - Pandemic Emergency Purchase Programme (PEPP): €1.85 trillion to stabilize markets.<br>- Ultra-Low Interest Rates: Further borrowing cost reductions.<br>- Eased Collateral Requirements: Allowed banks easier access to liquidity. |

### 4.2.2 Monetary Policy Tools and Implementation

The Eurosystem employs a range of monetary policy instruments to regulate liquidity, influence credit conditions, and ensure price stability across the euro area. Germany's monetary policy, shaped by the European Central Bank (ECB) and executed by the Bundesbank, evolved in response to macroeconomic developments, financial stability concerns, and inflationary trends. The key tools and their implementation are outlined below:

**Policy Interest Rates**

The ECB sets key policy rates that influence borrowing costs, credit activity, and overall economic conditions. These include:

- Main Refinancing Rate: Determines the cost of borrowing for commercial banks.
- Marginal Lending Facility Rate: Provides overnight liquidity to banks.

- Deposit Facility Rate: Influences interbank lending and liquidity conditions.

Between 2015 and 2019, the ECB maintained historically low interest rates to stimulate economic activity, with the deposit facility rate at -0.5% to encourage bank lending and investment. During the COVID-19 crisis (2020–2022), the ECB further reduced borrowing costs and implemented additional liquidity measures to sustain economic stability.

**Open Market Operations**

The Bundesbank, in coordination with the ECB, conducts various liquidity operations to manage monetary conditions:

- Main Refinancing Operations (MROs) & Longer-Term Refinancing Operations (LTROs): These provide liquidity to financial institutions, ensuring smooth credit flow.
- Targeted Longer-Term Refinancing Operations (TLTROs): Aimed at incentivizing bank lending to businesses and households, TLTROs were expanded in response to the pandemic to maintain credit supply and economic resilience.

**Minimum Reserve Requirements**

Banks operating in Germany must maintain minimum reserves with the Bundesbank, which affects short-term interest rates and liquidity conditions.

- To mitigate liquidity risks during the COVID-19 crisis, the ECB temporarily eased collateral requirements, allowing banks easier access to refinancing operations.

**Monetary Policy Asset Purchases**

The ECB implemented large-scale asset purchase programs to inject liquidity into financial markets and support credit expansion.

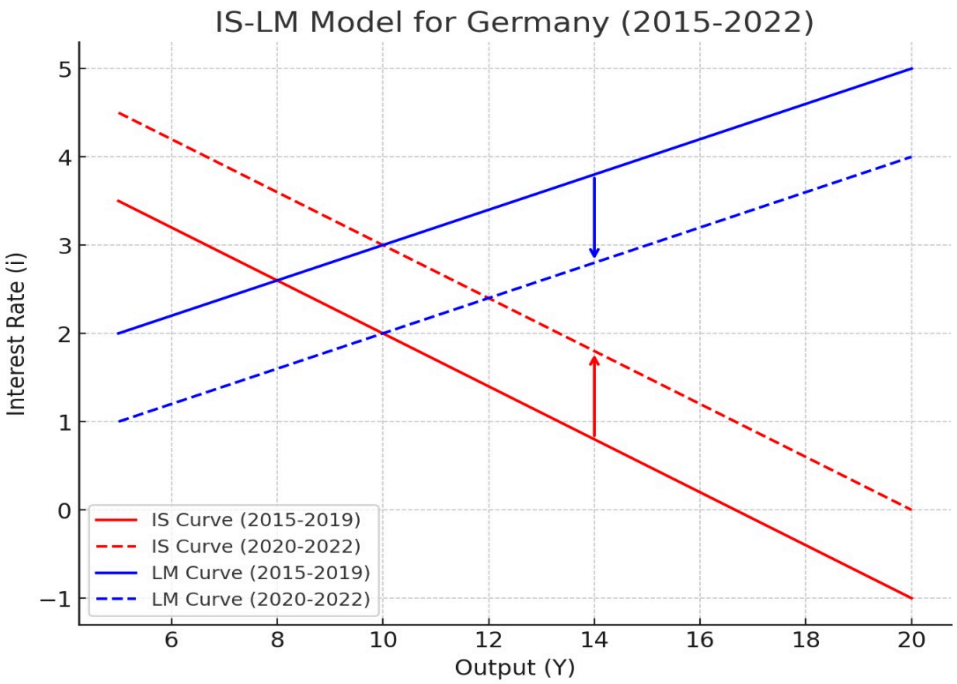
- Public Sector Purchase Programme (PSPP) (2015–2019): This quantitative easing measure aimed to stabilize inflation and promote economic growth.

- Pandemic Emergency Purchase Programme (PEPP) (2020–2022): In response to the economic shock of COVID-19, the ECB introduced PEPP with a total envelope of €1.85 trillion to stabilize financial markets and support economic recovery.

Through these monetary policy instruments, Germany navigated various economic challenges, ensuring financial stability while fostering economic recovery and growth.

# 5. IS-LM Analysis

In this paper, the IS-LM model will be used to illustrate the fiscal policy changes in Germany and their macroeconomic effects from a theoretical lens where the X-axis represents Germany's real GDP, typically measured in billion euros (adjusted for inflation) and the Y-axis represents interest rates (i), typically measured in percentages (%) (e.g., ECB policy rates or average lending rates).




During 2015-2019, Germany pursued expansionary monetary policy while maintaining a mildly contractionary fiscal policy (Schwarz Null). The LM curve shifted to the right, as the ECB maintained historically low interest rates, with the deposit facility rate at -0.5% to encourage lending. Open

market operations injected liquidity into the banking system, increasing money supply. The LM curve shifted downward (rightward), lowering interest rates while supporting investment and economic activity.

Germany maintained a budget surplus, steadily reducing the debt-to-GDP ratio. Public investment in infrastructure and social services was limited compared to aggressive fiscal expansion. While investment was supported by low interest rates, fiscal restraint kept the IS curve relatively stable with marginal rightward shifts due to public spending.

From 2020-2022, the economic impact of COVID-19 led to a drastic shift in both fiscal and monetary policies, driving substantial changes in the IS-LM equilibrium. The IS Curve therefore shifts rightward (Germany abandoned Schwarze Null and adopted expansionary fiscal policies). Government spending surged, with a €130 billion stimulus package (IEA, 2022), direct financial aid, and an increase in the Kurzarbeit Program. With a cut in VAT from 19% to 16%, the tax multiplier, combined with an increase in government expenditure and its interaction with the multiplier, would shift the IS curve significantly rightward, reflecting increased national income and aggregate demand, as well as an increase in interest rate which can be seen in the IS-LM diagram.

Likewise, the LM Curve Shifted Further Rightward as a result. The ECB introduced the €1.85 trillion PEPP, further increasing liquidity. Interest rates remained ultra-low, encouraging borrowing and investment. The LM curve shifted further downward, keeping interest rates suppressed. The combined impact of expansionary fiscal and monetary policy resulted in higher output (GDP) while preventing interest rates from rising sharply. However, the massive fiscal expansion also led to increased public deficits and rising debt-to-GDP levels, which may necessitate future fiscal adjustments.

# 6. Crowding In and Crowding Out effect

To study the crowding-in and crowding-out effects of fiscal policies in Germany, an analysis of the pre-pandemic (2015–2019) and post-pandemic (2020–2022) periods is necessary:

**Crowding-In Effects**

- Pre-Pandemic (2015–2019):
  Germany's balanced budget policy (Schwarz Null) and low ECB interest rates created a favorable investment environment. Public investments in digital infrastructure, renewable energy, and refugee integration had a strong crowding-in effect, as they complemented rather than displaced private investment. For example, government support for green energy initiatives encouraged private firms to expand into renewable technologies, leading to increased investment in wind and solar energy sectors. Additionally, targeted fiscal incentives in the technology sector promoted private R&D, fostering long-term industrial innovation.
- Post-Pandemic (2020–2022):
  The large-scale fiscal response to the COVID-19 crisis—especially the Kurzarbeit wage subsidy program, direct business grants, and healthcare investments—stabilized the private sector and prevented a deeper economic collapse. These interventions effectively crowded in investment in technology, healthcare, and renewable energy, where firms took advantage of public incentives to expand operations. For example, increased funding for vaccine research and healthcare infrastructure enabled pharmaceutical companies and biotech firms to scale up investments. Additionally, government subsidies for green energy projects encouraged private sector participation in Germany's energy transition.

**Crowding-Out Effects**

- Pre-Pandemic (2015–2019):
  While Germany's fiscal restraint kept public debt low, limited public infrastructure spending restricted the full potential of private investment. Economists argue that underinvestment in transport, broadband infrastructure, and education resulted in foregone private sector growth, particularly in capital-intensive industries that relied on modern infrastructure for efficiency. This reflects a form of passive crowding-out, where insufficient government investment failed to catalyze further private sector expansion.
- Post-Pandemic (2020–2022):
  Initially, fiscal stimulus did not lead to significant crowding-out, as the ECB's expansionary policies maintained low borrowing costs. However, by 2022, the situation shifted as inflationary pressures led the ECB to tighten monetary policy, raising interest rates and

reducing liquidity. This resulted in higher borrowing costs for businesses, particularly in capital-intensive and energy-intensive sectors.

Germany's fiscal policies largely resulted in crowding-in effects in technology, healthcare, and renewable energy, where government interventions complemented private sector investment. However, post-pandemic monetary tightening and energy price volatility introduced crowding-out effects, particularly in energy-intensive manufacturing, construction, and SMEs. While Germany successfully leveraged fiscal tools to stimulate key sectors, long-term growth depends on striking a balance between public investment and fiscal discipline.

| **Aspect** | **Pre pandemic (2015-2019)** | **Post pandemic (2020-present)** |
|---|---|---|
| Debt to GDP ratio | Declined (71-59%) | Increased to 70% |
| Bond Yields | Low or negative | Low during pandemic rising post 2022 |
| Private investment | Moderate growth | Sharp contraction in 2020, recovery in 2021–2022 |
| Fiscal policy stance | Restrained, focused on targeted investments | Expansionary, with massive stimulus |
| Crowding out risk | Low (due to low interest rates and fiscal discipline) | Moderate (rising interest rates post-2022) |
| Crowding in potential | Limited by fiscal restraint | High(due to stimulus and infrastructure spending) |

# 7. Impact on Private Investment

The impact of fiscal and monetary policies on private investment in Germany post-2015 can be analyzed across the pre-pandemic (2015–2019) and post-pandemic (2020-2022) segments. During each time frame private investment levels responded differently to the unique policy responses and economic conditions that existed.

**1. Pre-Pandemic (2015–2019): Moderate Growth and Fiscal Restraint**

Germany's fiscal policy maintained a balanced budget by preventing new debt creation and limiting public spending, ensuring macroeconomic stability but constraining large-scale public investments. However, targeted investments in digital infrastructure, renewable energy, and refugee integration fostered private sector growth through a crowding-in effect. Moderate tax relief measures further stimulated private investment by increasing disposable income and corporate profitability.

On the monetary side, the ECB's accommodative stance kept interest rates low and implemented quantitative easing to support borrowing. However, negative interest rates, while encouraging lending, also squeezed bank profit margins, potentially restricting their capacity to finance private investment.

During this period, private investment expanded at a moderate rate due to reduced borrowing costs and strong export demand, though it can be argued that Germany's limited public spending and insufficient infrastructure investment constrained its full potential.

**2. Post-Pandemic (2020–2022): Sharp Contraction and Fiscal Expansion**

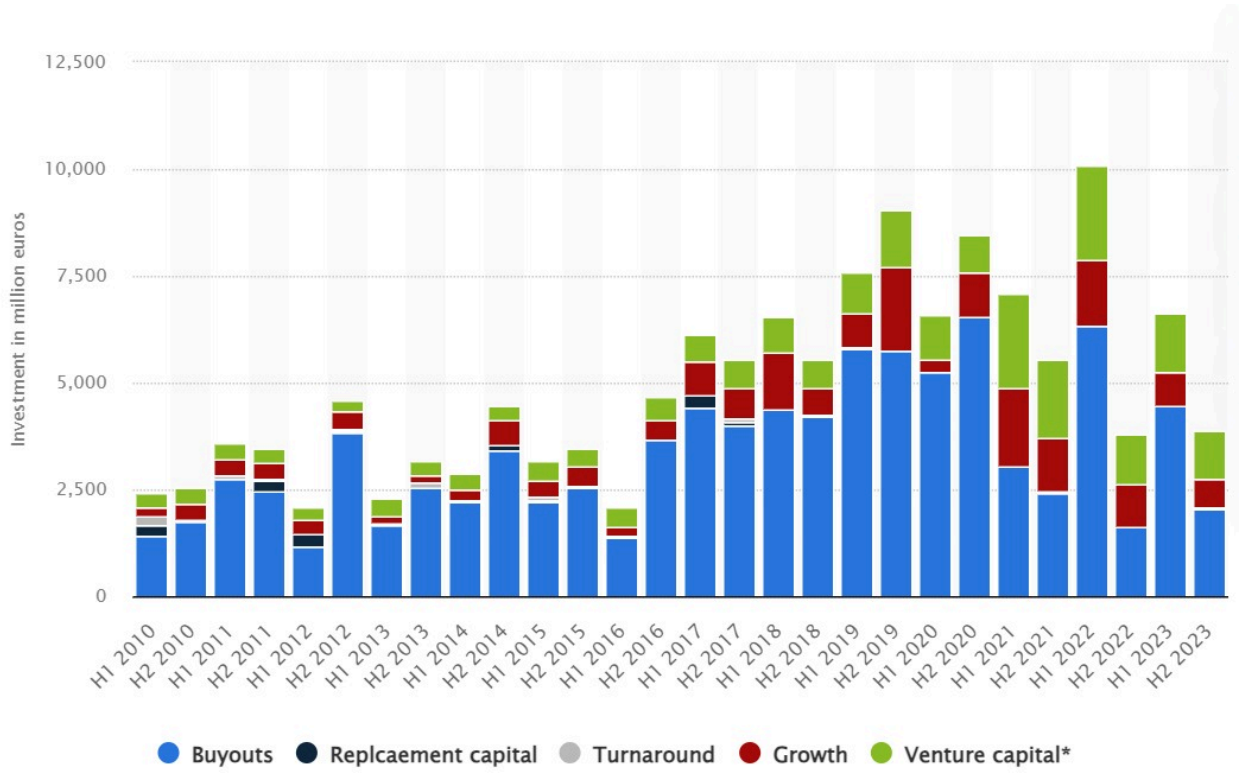
From 2020 to 2021, Germany suspended its debt brake and implemented massive stimulus packages to counteract the economic downturn. Key fiscal measures included the Kurzarbeit Scheme to protect jobs and household incomes, direct business support through grants and loans, and increased healthcare and vaccine spending to strengthen the public health response. A temporary VAT reduction was also introduced to boost consumer spending and support businesses.

On the monetary front, the ECB launched the Pandemic Emergency Purchase Program (PEPP) to maintain market liquidity and financial stability while keeping ultra-low interest rates to encourage borrowing and investment.

While private investment initially contracted sharply due to lockdowns and uncertainty, it rebounded by late 2020, with growth accelerating in technology, healthcare, and green energy sectors, supported by the Kurzarbeit scheme, which preserved human capital and facilitated a smoother recovery.

By 2022, rising inflation and the energy crisis prompted a shift in Germany's policy stance. Fiscal measures included energy price caps to protect consumers and businesses from soaring costs and

increased investments in renewable energy to reduce reliance on Russian gas. However, the reinstatement of the debt brake limited further stimulus spending. On the monetary side, the ECB tightened its policies by raising interest rates and scaling back asset purchases, leading to higher borrowing costs and reduced liquidity. These measures created challenges for private investment, particularly in energy-intensive industries, while green energy, digital transformation, and infrastructure investments remained resilient due to structural trends and public sector incentives.




*Source: Statista*

## 8. Conclusion & Policy Recommendations

Germany has maintained a complex policy mix since 2015 that combines fiscal restraint with specific financial measures to influence private investment results and economic growth in different ways. Germany's adherence to a balanced budget policy known as Schwarz Null before the pandemic and

low interest rates from the ECB's monetary policy prevented crowding-out effects yet restricted opportunities for increased public investment to stimulate economic growth. The moderate growth of private investment benefited from strong export demand and low borrowing costs while infrastructure underinvestment limited potential productivity improvements.

Germany's extensive pandemic stimulus measures, which included Kurzarbeit, subsidies, and direct business support, proved essential for economic stabilization and helped avoid further drops in private investment. The ECB's expansionary monetary policy contributed to the economic recovery by keeping borrowing costs low. Government interventions succeeded in attracting private investment towards healthcare, green energy and digital transformation fields.

Since the pandemic recovery period began, private investment faces new challenges from rising interest rates and increased energy costs which particularly impact industries that consume a lot of energy. Digital infrastructure benefited from supportive fiscal policies but future crowding-in effects face potential limitations from renewed fiscal discipline and tighter monetary measures.

**Proposed policy measures:**

- Germany needs to contemplate various strategies to improve future fiscal-monetary coordination.
- Targeted Public Investments: Private sector productivity can be boosted through strategic investments in infrastructure and innovative technologies.
- Flexible Fiscal Rules: Modify the debt brake system to enable counter-cyclical expenditure during periods of economic crisis.
- Energy Transition Support: Maintain continuous financial support for both clean energy initiatives and technological advancement.
- Monetary-Fiscal Synergy: The ECB policies must be synchronized with national fiscal goals to ensure advantageous borrowing conditions.

The implementation of these strategies will enable Germany to achieve fiscal sustainability alongside the stimulation of private investment and long-term economic growth.

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